Hospital-Acquired Influenza Tables

Dictionary and Schema Guide

A readable reference for both medical and technical collaborators

What this document is for

This guide explains the tables that organize the evidence base for a hospital-acquired influenza system. It translates each column into plain language, lists its expected values, and explains why that field matters for risk modeling, data engineering, and operational decision support.

Quick mental model

- Table 1 stores factors linked to acquiring influenza inside the hospital.
- Table 2 stores severity or outcome factors after influenza is already present.
- Secondary infection tables store what happens after influenza when other infections appear.
- Policy tables store “what to do” rules; they should not be mixed with statistical risk factors.
- A core design goal is to prevent leakage: the model should not use future outcome information to predict earlier risk.

1. Table map: what each table is for

Table	Main question	Good content	Should not be mixed in
nosocomial_influenza_risk_factors	What increases or decreases the chance of catching influenza in the hospital?	Baseline vulnerability, early clinical signals, room and ward features, exposure variables, protective factors, surveillance logic.	ICU need, mortality, ARDS, downstream secondary infection outcomes.
hospitalized_influenza_severity_factors	Once a patient has influenza, what is linked to worse outcomes?	Mortality, ICU admission, ventilation need, complication severity, comparative outcome markers.	Pre-infection acquisition predictors.
secondary_infections_in_influenza	After influenza is present, what predicts a secondary hospital infection?	Mechanical ventilation, sepsis, immune depletion, age, anemia, late complications.	Acquisition of influenza itself.
infection_control_policies (companion table)	When risk is recognized, what actions are allowed or recommended?	Isolation, PPE, prophylaxis, testing, visitor rules, staff restriction, ventilation measures.	Empirical odds ratios or ML feature importance.

Engineering analogy: Table 1 is like pre-event feature engineering, Table 2 is post-event severity analytics, secondary infection tables are downstream event models, and policy tables are the action layer that sits on top of model outputs.

2. Shared schema dictionary

How to read the dictionary: For each column, this guide gives a plain-language definition, typical values, and why the field matters to a technical or operational reader.

2.1 Source and provenance columns

source_id

Plain-language meaning: Stable internal identifier for the paper or source record.

Typical values / examples: Short machine-friendly strings such as zhang_2025 or godoy_2020.

Why a technical user should care: Acts like a primary key for joins, deduplication, traceability, and citation back to the paper.

citation

Plain-language meaning: Human-readable short reference for the study.

Typical values / examples: Examples: “Yang et al. 2020”, “Cho et al. 2024”.

Why a technical user should care: Best for dashboards, reports, and fast manual review. It is not ideal as a unique key by itself.

pmid

Plain-language meaning: PubMed identifier for the article.

Typical values / examples: Numeric PubMed IDs, for example 33213361; can be NR when unavailable.

Why a technical user should care: Useful for automatic retrieval, enrichment, and linking to biomedical literature tools.

doi

Plain-language meaning: Digital Object Identifier for the paper.

Typical values / examples: Strings like 10.1186/s12879-020-05580-9.

Why a technical user should care: Useful for stable cross-system linking and article retrieval even outside PubMed.

country

Plain-language meaning: Country where the study took place.

Typical values / examples: China, France, Spain, USA, Sweden, etc.

Why a technical user should care: Helps with external validity: hospital structure, vaccination policies, room models, and practice patterns vary by country.

setting

Plain-language meaning: Short description of the hospital environment studied.

Typical values / examples: Tertiary hospital, ICU network, oncology unit, university hospital outbreak setting, surveillance catchment.

Why a technical user should care: Important for deciding whether a feature is portable to another hospital environment.

population

Plain-language meaning: Who was actually studied.

Typical values / examples: Adults hospitalized >7 days, ICU patients, oncology patients, ward-level observations, healthcare workers.

Why a technical user should care: Defines the denominator for interpretation. A predictor in oncology patients may not generalize to a general ward.

study_design

Plain-language meaning: How the evidence was generated.

Typical values / examples: Matched case-control, observational cohort, retrospective ML study, ecological study, Bayesian transmission model.

Why a technical user should care: Critical for evidence weighting: adjusted observational evidence is interpreted differently from descriptive or ML feature-importance evidence.

2.2 Clinical and epidemiologic context

pathogen_scope

Plain-language meaning: Which pathogen family the row applies to.

Typical values / examples: influenza_only, influenza A, influenza + other respiratory viruses, shared_with_other_pathogens.

Why a technical user should care: Prevents accidental overgeneralization. A feature from mixed viral respiratory infections may be only supportive for influenza.

acquisition_context

Plain-language meaning: Whether the row is about hospital-acquired influenza, a comparison group, or transmission context.

Typical values / examples: Hospital-acquired, hospital-acquired / compared, healthcare-associated / expanded exposure definition, hospital-acquired transmission context.

Why a technical user should care: Separates direct acquisition evidence from comparative or surveillance logic.

hospital_acquired_definition

Plain-language meaning: How the study defined hospital acquisition.

Typical values / examples: >48 h after admission, >=72 h after admission, >=7 days after admission, positive PCR after hospital day 3, etc.

Why a technical user should care: Very important for label quality. Different definitions can change who is counted as hospital-acquired.

diagnostic_method

Plain-language meaning: How influenza or the relevant exposure state was measured.

Typical values / examples: RT-PCR, PCR + ML, institutional surveillance, sequencing, microbiological confirmation.

Why a technical user should care: Helps assess reliability and feasibility. PCR-based rows are usually stronger than vague syndromic definitions.

surveillance_frame

Plain-language meaning: Time window or season covered by the study.

Typical values / examples: 2011–2020, 2018–2019, 2004–2007 outbreaks, six influenza seasons.

Why a technical user should care: Important for context, especially when policies, testing practices, or dominant subtypes changed over time.

comparator_group

Plain-language meaning: What the row is being compared against.

Typical values / examples: HA vs non-HA, nosocomial vs community-acquired, cases vs controls, ward comparison, transmission-source modeling.

Why a technical user should care: Lets a technical reader understand whether the row is causal, comparative, or descriptive.

clinical_context

Plain-language meaning: Short explanation of the care context where the row applies.

Typical values / examples: Adult inpatients, severe influenza, oncology, hospital outbreaks, population-based surveillance.

Why a technical user should care: Useful for filtering rows to the closest match for deployment.

applicability_to_influenza

Plain-language meaning: How directly the row applies to influenza.

Typical values / examples: influenza_only, shared_with_other_pathogens, broader respiratory-virus context.

Why a technical user should care: A row can still be useful even if not influenza-only, but it should usually be marked as supportive rather than core.

2.3 Feature and variable semantics

canonical_feature_name

Plain-language meaning: The normalized name of the feature after synonyms are merged.

Typical values / examples: age, lymphopenia, room_sharing, patient_exposure, hcw_exposure, influenza_vaccination.

Why a technical user should care: This is the main machine-friendly field name. It prevents the same concept from appearing under many aliases.

risk_factor_name (optional legacy field)

Plain-language meaning: Original paper-specific label before normalization.

Typical values / examples: lymphocytopenia, chronic_renal_failure, double_or_multi_occupancy_room.

Why a technical user should care: Useful during extraction audits, but canonical_feature_name should drive modeling.

risk_factor_type

Plain-language meaning: What kind of evidence unit the row represents.

Typical values / examples: clinical_association, model_feature, attributable, descriptive, surveillance_classification, comparative_outcome.

Why a technical user should care: Tells you whether the row is a statistical association, a machine-learning feature, a modeled attributable fraction, or only descriptive context.

feature_class

Plain-language meaning: High-level category of the feature.

Typical values / examples: patient, comorbidity, lab, radiology, exposure, structural, operational, surveillance, staff_vector, pathogen, diagnostic.

Why a technical user should care: Acts like a typed schema layer. Very useful for filtering, validation, and downstream UI grouping.

temporal_stage

Plain-language meaning: When the feature exists relative to infection or outbreak evolution.

Typical values / examples: baseline, early_hospitalization, during_exposure_window, outbreak_dynamic, ward_level_static, surveillance_classification.

Why a technical user should care: One of the most important leakage-protection columns. If the value happens after infection, it should not be used in an acquisition model.

risk_factor_definition

Plain-language meaning: Human-readable explanation of what the feature means in the source study.

Typical values / examples: “Admission to a geriatric ward”, “Low lymphocyte count”, “Exposure to recorded contagious healthcare workers”.

Why a technical user should care: Important when a canonical feature name is too compact to interpret on its own.

finding_text

Plain-language meaning: Short plain-language statement of the study result.

Typical values / examples: “Associated with HA influenza”, “Most influential predictor in RF model”, “Major transmission source”.

Why a technical user should care: This is usually the fastest field for manual review.

2.4 Statistical evidence columns

effect_size

Plain-language meaning: Numeric size of the reported effect, when available.

Typical values / examples: 3.11, 0.42, 59%, 8.0, NR.

Why a technical user should care: Do not assume the same meaning across rows; always read together with effect_size_type.

effect_size_type

Plain-language meaning: What the effect size represents.

Typical values / examples: OR, aOR, HR, IRR, proportion, attributable, model feature, cluster count, sensitivity / specificity.

Why a technical user should care: Essential for correct interpretation. An odds ratio and a descriptive proportion are not interchangeable.

confidence_interval

Plain-language meaning: Uncertainty interval for the effect size.

Typical values / examples: 1.24–7.80, 0.23–0.78, 34–83 (CrI), NR.

Why a technical user should care: Technical users can use this to judge precision; medical users can read it as “how stable or uncertain the estimate is.”

p_value

Plain-language meaning: Reported probability measure for hypothesis testing, when available.

Typical values / examples: <0.001, 0.03, NR.

Why a technical user should care: Helpful but should not be the only evidence filter. A clinically important signal can still be useful even if the paper is descriptive.

effect_direction

Plain-language meaning: Direction of the association or interpretation.

Typical values / examples: increase_risk, decrease_risk, contextual, mixed_or_unclear.

Why a technical user should care: Turns a result into a machine-friendly sign. It is one of the easiest columns to use in dashboards or rule engines.

evidence_strength

Plain-language meaning: Practical confidence rating assigned to the row.

Typical values / examples: high, moderate, medium, low, contextual.

Why a technical user should care: Not a formal GRADE score; it is a working operational rating for prioritization.

adjustment_level

Plain-language meaning: How much statistical adjustment the study used.

Typical values / examples: Multivariable logistic regression, adjusted odds ratio, ML feature importance, descriptive comparison, Bayesian model.

Why a technical user should care: Very important for engineers: this is a quick proxy for confounding control and evidence robustness.

outcome_linked

Plain-language meaning: What the factor is linked to in the source study.

Typical values / examples: acquisition, transmission, mortality, secondary infection, acquisition classification.

Why a technical user should care: Helps maintain separation between “getting influenza” and “what happens after influenza.”

2.5 Interpretation and implementation columns

key_limitations

Plain-language meaning: Short note about the major weaknesses of the evidence.

Typical values / examples: Single center, case-case design, outbreak-specific, abstract-only data, no per-factor OR reported.

Why a technical user should care: A technical reader can treat this as a caution flag before overfitting to one paper.

study_reported_implication

Plain-language meaning: What the paper itself suggests the finding means.

Typical values / examples: “Room sharing increases transmission”, “Vaccination was protective”, “Longer stay increases exposure opportunity”.

Why a technical user should care: Useful when converting paper language into a system-facing design.

codeblue_translation

Plain-language meaning: Operational translation into system logic.

Typical values / examples: Add room occupancy feature; add patient vaccination as protective factor; use recent discharge flag.

Why a technical user should care: This is where biomedical evidence is turned into feature engineering or decision-support language.

tags

Plain-language meaning: Short free-text labels for filtering or UI grouping.

Typical values / examples: RISK PROFILE, LAB SIGNAL, STRUCTURAL RISK, PREVENTION, WORKER VECTOR.

Why a technical user should care: A lightweight navigation layer; good for search, filters, and quick human scanning.

factor_role

Plain-language meaning: The semantic role of the row inside the system.

Typical values / examples: acquisition, protective, transmission_context, severity_associated, comparative_profile.

Why a technical user should care: A high-level routing field that says where the row belongs conceptually.

data_status

Plain-language meaning: Readiness and curation state of the row.

Typical values / examples: ready, ready_partial, draft.

Why a technical user should care: Helps decide what can be used immediately versus what still needs verification.

3. Controlled vocabularies and recommended values

3.1 feature_class

Value	Meaning	Examples
patient	General patient attribute not better classified elsewhere.	age, BMI, pregnancy status
comorbidity	Pre-existing disease burden.	chronic_kidney_disease, malignancy, immunodeficiency
lab	Laboratory measurement or derived blood parameter.	lymphopenia, hypoalbuminemia, platelet count, DNI
radiology	Imaging-based signal.	abnormal chest X-ray, pleural effusion, pneumonia_at_admission
exposure	Direct or indirect opportunity to catch infection.	patient_exposure, hospital_exposure_duration, roommate exposure
structural	Built environment or room layout variable.	room_sharing, double room, ward density
operational	Workflow, policy, or hospital process variable.	mask_use_intensity, vaccinate_or_mask_policy
staff_vector	Staff-related transmission or prevention variable.	hcw_exposure, hcw_vaccination_rate
surveillance	Case-definition or healthcare-exposure logic.	recent discharge exposure, expanded HAI definition
pathogen	Pathogen or subtype context rather than patient feature.	H3N2-associated dissemination
diagnostic	Measurement or investigation capability.	rapid testing performance, WGS-confirmed cluster

3.2 temporal_stage

Value	Meaning	Why it matters
baseline	Known before the hospital risk window starts.	Safest for prediction; usually low leakage risk.
early_hospitalization	Observed early enough to still support prevention.	Useful for near-real-time risk updating.
during_exposure_window	Active during the period in which exposure could lead to infection.	Very important for dynamic risk models.
outbreak_dynamic	Changes as an outbreak unfolds.	Useful for transmission monitoring and escalation logic.
ward_level_static	More stable ward or room characteristic.	Good for risk stratification and simulation.
surveillance_classification	Definition or labeling logic	Should not be confused with

	rather than biology.	causal patient risk.
post_infection	Occurs after influenza is already present.	Should be excluded from Table 1 acquisition models.

3.3 effect_direction

Value	Meaning	Typical interpretation
increase_risk	Higher values or presence raise risk.	Positive predictor for acquisition or worsening.
decrease_risk	Higher values or presence lower risk.	Protective factor.
contextual	Important for context, but not a simple up/down biological signal.	Burden estimates, surveillance logic, outbreak structure.
mixed_or_unclear	Direction is not clean or is dependent on context.	Needs careful manual handling.

3.4 evidence_strength

Value	Meaning	Typical source
high	Strong adjusted evidence or repeated strong support.	Multivariable studies with clear effects.
moderate	Useful signal, but with notable limits.	Univariate findings, ML importance without ORs, single-context evidence.
low	Weak or incomplete evidence.	Abstract-only feature importance, sparse data.
contextual	Not a direct predictor but still important.	Transmission-attributable fractions, surveillance definitions.

3.5 factor_role

Value	Meaning	Where it usually belongs
acquisition	Core predictor of getting influenza in the hospital.	Table 1
protective	Reduces acquisition risk or exposure risk.	Table 1 or policy-aware companion views
transmission_context	Describes how spread happens or how it is recognized.	Table 1 supportive layer / transmission support
severity_associated	Linked to worse outcomes after infection.	Table 2
comparative_profile	Describes differences between groups, not always a deployable predictor.	Often Table 2 or supplementary context

3.6 data_status

Value	Meaning	When to use

ready	Fully curated and acceptable for downstream use.	May be used in analysis or modeling.
ready_partial	Mostly usable, but some detail still needs confirmation.	Safe for review; use cautiously in modeling.
draft	Still provisional.	Do not treat as finalized evidence.

3.7 applicability_to_influenza

Value	Meaning	Operational recommendation
influenza_only	Directly about influenza.	Best for core evidence.
shared_with_other_pathogens	Mixed respiratory-virus context but still relevant.	Usually supportive, not primary.
broader respiratory-virus context	General respiratory hospital transmission logic.	Good for policy or transmission support, not always for primary weights.

4. Reading rules for mixed medical + technical teams

- Do not mix acquisition predictors with severity outcomes. A factor like ICU admission may be very important, but it usually belongs after infection, not before it.
- Always read effect_size together with effect_size_type. “3.1” means very different things if it is an OR versus a cluster count.
- Canonical names should drive modeling. Original paper wording is useful for auditing, but not for production schema design.
- Rows marked contextual can still matter operationally. For example, an expanded healthcare-associated definition can improve surveillance even if it is not a biological predictor.
- A single paper can contribute to more than one table, but each row must be placed according to its timing and semantic role.
- The policy layer is separate on purpose. The evidence tables answer “what seems risky?” while the policy table answers “what are we allowed to do about it?”.

Engineering shortcut

- source_id = provenance key
- canonical_feature_name = normalized feature name
- feature_class = feature type
- temporal_stage = anti-leakage guardrail
- effect_direction + evidence_strength = prioritization signal
- factor_role = routing field to the correct table or UI view
- codeblue_translation = bridge from literature to system implementation

Medical shortcut

- The schema separates who is vulnerable, who was exposed, how infection may have spread, and what happened after infection.
- If a field happens after influenza is diagnosed, it should usually not be used to predict acquisition of influenza.
- Protective factors such as vaccination or safer ward structures should stay visible rather than being treated as the absence of risk.

5. Recommended companion table for policies and recommendations

Why this matters: WHO guidance, CDC guidance, and hospital protocols are important, but they are not the same kind of evidence as odds ratios or ML features. A separate policy table lets the model output risk, while the policy layer translates risk into allowed actions.

organization

Plain-language meaning: Source of the rule.

Typical values / examples: WHO, CDC, national ministry, local hospital.

Why a technical user should care: Needed for versioning, trust, and local override logic.

policy_name

Plain-language meaning: Short title of the recommendation.

Typical values / examples: Droplet precautions, oseltamivir prophylaxis, visitor restriction, isolation after positive PCR.

Why a technical user should care: Human-readable label for governance and UI.

policy_type

Plain-language meaning: Type of intervention.

Typical values / examples: isolation, PPE, testing, prophylaxis, ventilation, staffing, visitor control.

Why a technical user should care: Makes policies searchable and machine-routable.

trigger_condition

Plain-language meaning: When the policy should activate.

Typical values / examples: positive test, high risk score, same-room exposure, ward outbreak threshold.

Why a technical user should care: This is the bridge from evidence tables to action rules.

recommended_action

Plain-language meaning: What the system or staff should do.

Typical values / examples: isolate patient, cohort, test exposed staff, reduce rotation, start prophylaxis per protocol.

Why a technical user should care: This is the operational output layer.

evidence_level

Plain-language meaning: How strong or authoritative the recommendation is.

Typical values / examples: guideline strong recommendation, local policy, expert consensus.

Why a technical user should care: Keeps recommendations transparent and reviewable.

operational_feasibility

Plain-language meaning: How realistic the action is in a given hospital.

Typical values / examples: high, medium, low; or capacity-dependent.

Why a technical user should care: Critical for real-world deployment: a good rule that cannot be executed should be flagged as such.

policy_version

Plain-language meaning: Version or date for the policy.

Typical values / examples: WHO_2024_v1, Hospital_X_2026_Q1.

Why a technical user should care: Needed for audit trails and future updates.

One-line summary: the evidence tables tell you what is risky; the policy table tells you what the hospital is allowed or expected to do in response.

Prepared as a joint reference for clinical, research, and engineering collaborators.